Assignment - 9

WAP to implement first come first served (FCFS) CPU scheduling for 5 processes, accept arrival time, burst time for each process and calculate the waiting time and turn around time and find the average of waiting time and turnaround time.

PID	AT	BT
P1	4	5
P2	0	3
P3	8	6
P4	3	4
P5	5	2
P6	12	8

```
#include <stdio.h>
int main()
{
printf("Enter number of processes:");
scanf("%d",&n);

int at[n], bt[n], ct[n], tat[n], wt[n], rt[n];
for(i=0; i<n; i++)
{
printf("Enter AT and BT for P%d :", i+1);
scanf("%d %d", &at[i], &bt[i]);
ct[0] = at[0] + bt[0];
for(i=1; i<n; i++)
{
if(ct[i-1] < at[i])
ct[i] = at[i] + bt[i];

else ct[i] = ct[i-1] + bt[i];
}

float avg_tat = 0, avg_wt = 0;
```

```
for(i=0; i<n; i++)
{
tat[i] = ct[i] - at[i];
wt[i] = tat[i] - bt[i];
rt[i] = wt[i];
avg_tat += tat[i];
avg_wt += wt[i];
}

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\tRT\n");
for(i=0; i<n; i++)
printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
i+1, at[i], bt[i], ct[i], tat[i], wt[i],
rt[i]);

printf("\nAverage TAT = %.2f", avg_tat/n);
printf("\nAverage WT = %.2f", avg_wt/n);

return 0;
}
```

OUTPUT

```
Enter number of processes: 6
Enter Arrival Time and Burst Time:6s
P1 AT : 4
P1 BT : 5
P2 AT : 0
P2 BT : 3
P3 AT : 8
P3 BT : 6
P4 AT : 3
P4 BT : 4
P5 AT : 5
P5 BT : 2
P6 AT : 12
P6 BT : 8

PID AT BT CT TAT WT RT
P2 0 3 3 3 0 0
P4 3 4 7 4 0 0
P1 4 5 12 8 3 3
P5 5 2 14 9 4 4
P3 8 6 20 12 6 6
P6 12 8 28 16 8 8

Average turn around time = 7.00
Average waiting time = 4.00
```